SDG Blockchain Accelerator Cohort 2
Deliverables Roadmap

Accelerator Duration: 29 September - 15 December 2025
Challenge Owner (CO): UINDP Bosnia and Herzegovina
Solution Maker (SM): SALA

1. Challenge Definition

1.1 Refined Challenge Statement

*Bosnia and Herzegovina's fragmented higher-education governance and manual diploma checks slow mobility and enable fraud. We will pilot **EU-standards aligned digital diplomas** that students control in a wallet and employers can verify in seconds via a public portal, with only a tamper-proof fingerprint anchored on Cardano testnet (no PII on-chain). This complements paper diplomas and informs Sarajevo Canton legal alignment.*

1.2 Local Context

Manual verification adds delays/costs and undermines trust, especially for cross-border study and employment. Primary beneficiaries are University of Sarajevo students/graduates; secondary beneficiaries include employers, registrar offices, and public agencies (HEA, CIP, Sarajevo Canton). UTIC centralizes university IT and diploma production and is a key implementation partner.

1.3 Alignment with UNDP Country Office and National Development Priorities

Directly advances the UNDP Country Programme (2026–2030) on accountable institutions and digital governance; UNDP CO provides technical assistance and convening. University of Sarajevo co-finances/hosts; solution aligns with EU open standards and future national digital wallet (IDDEEA) integration.

1.4 Relevance to UNDP Programmes and Resource Mapping

New initiative under University leadership with UNDP Accelerator Lab support. University provides technical/legal collaboration and hosting; UNDP contributes in-kind product management, standards alignment, and stakeholder convening.

1.5 Key Stakeholders & Potential Partners

University of Sarajevo (Rectorate, UTIC, pilot faculties), UNDP CO; public stakeholders: Ministry of Civil Affairs, Sarajevo Canton ministries, HEA, CIP, IDDEEA, Data Protection Agency; employers' associations, student unions; tech/standards partners: ***Emurgo Labs/Cardano*** *and* ***Sala***

1.6 Expected Impact

Trusted, fast diploma verification through an EU-aligned decentralized trust framework; empowers graduates, strengthens integrity without centralizing control, informs policy updates/SOPs, and positions the solution to scale nationally (IDDEEA/EUDI) or as a Digital Public Good.

1.7 Target SDGs and Indicators

- SDG 16 (Effective, accountable institutions) – user satisfaction with diploma services (16.6.2).
- SDG 4 (Quality education) – improved recognition/verification processes (national context).

2. User Mapping & Testing Strategy

2.1 Primary User Persona

Recent University of Sarajevo graduate preparing job/masters applications; digitally savvy; needs fast, privacy-preserving proof without translation/notary/apostille; wants to reuse a digital credential with selective disclosure.

2.2 User Story

- As a *recent graduate/student*, I want to *receive my digital diploma (and course certificates) into my digital wallet as soon as I graduate*, so that *I always have a trusted proof of my qualifications*.
    - After graduation, I receive an email from the registry office with a secure link/QR to claim my credential.
    - I can add the credential to a standards-compliant wallet, following step-by-step instructions.
    - The credential shows the issuer (University of Sarajevo), credential type, issue date, and a summary of courses completed.
    - No personal data is stored on-chain; only a document "fingerprint" is anchored for integrity.
    -
- As a *recent graduate/student*, I want to *share my credentials presentation* so that *employers or universities can verify them in seconds without contacting the University*.
    - A verifier who scans the QR or opens the link sees a validation page showing "Valid" status, issuer (University of Sarajevo), credential type, and issue date.

    - The validation page is available in at least two EU languages (English and German), so I can use it cross-border.
- As a *recent graduate/student*, I want to *disclose only the minimum information required (i.e., degree title and year, without grades)*, so that *I protect my privacy, while meeting specific application requirements*.
    - I can select which fields to disclose, with the degree title and date being the minimum.
    - I can download a QR-coded PDF that, when scanned, triggers the same verification flow and shows the verifier (potential employer, higher education institution) the up-to-date status.
- As a *registrar's office administrator*, I want to *issue diplomas to students wallets, with their "fingerprint" anchored on the blockchain*, so that *students' records are tamper-proof and instantly verifiable*.
- As an *employer*, I want *to scan a QR code or upload a certificate link*, so that *I can instantly verify its authenticity without waiting weeks for manual checks*.
- As an *educational policy maker*, I want *improved diploma issuance reporting* across universities, so that *I can track systemic risks and report on education integrity*.
-

2.3 Testing Strategy

Weeks **7–8**: structured tests with students, registrars, and employers; collect feedback in a pilot review workshop; iterate UI/flows.

2.4 Feedback Loop Design

UNDP user research lead coordinates discovery, usability tests, accessibility (WCAG), and translates insights into UX requirements; analytics/KPIs inform iterations.

3. Solution Overview

3.1 Solution Summary

Digital diploma issuance & verification platform for University of Sarajevo, ***anchored on Cardano testnet****, aligned with* ***W3C VC*** *and* ***OpenID4VCI/VP****; MVP demonstrates issuance → wallet → verification and provides inputs for legal alignment.*

3.2 Core Features and Functionalities

What are the most important features of your solution?

- **Registrar Admin:** CSV/Excel upload → bulk issuance (W3C VC v2.0, OID4VCI).
- **Wallet Delivery:** students receive and present via OID4VP (QR).
- **Verifier Portal:** public, instant **Valid/Revoked** with issuer, credential type, issue date.
- **Revocation/Status:** registrar revokes; verifier reflects status

3.3 Tech Stack and Tools

- ***Standards:*** *W3C VC v2.0; OID4VCI/VP; selective disclosure.*
- ***Issuer/Verifier:*** *Web (Vue.js);* ***API:*** *Nest.js;* ***DB:*** *PostgreSQL (metadata);* ***Admin:*** *Vue.js.*
- ***Wallet:*** *standards-compatible demo wallet (OID4VCI-enabled).*
- ***Blockchain: Cardano testnet*** *for integrity proof & status anchoring.*

3.4 Blockchain Functionality

What blockchain-specific interactions are included (e.g., token minting, smart contracts, on-chain verification)?

- *On-chain* ***hash/fingerprint anchoring*** *of issued credentials (no PII).*
- *Simple* ***revocation/status*** *registry reflected in verifier results; future multi-chain (Stellar, FLock) for performance/analytics.*

4. Weekly Workplan

Briefly describe your team's focus during each stage of the program. (You may use the workplan template from your hackathon roadmap.)

- ***W1–2:*** *Scope, personas, KPIs; choose wallet; approve simple screens.*
- ***W3–4:*** *First* ***on-chain hash*** *anchored; first VC issued to demo wallet.*
- ***W5:*** *Verifier portal minimum functions online.*
- ***W6:*** *Registrar admin minimum functions online.*
- ***W7–8:*** *Stakeholder testing; collect feedback.*
- ***W9–10:*** *Legal alignment & standards note (Sarajevo Canton).*
- ***W11:*** *Integrate issuer + wallet + verifier into one MVP flow.*
- ***W12:*** *Pilot demo with 3 use cases (job, mobility, employer check).*

5. Final Deliverables

5.1 Solution Demo (PoC)

A working or visual prototype that demonstrates the core idea. (To be scheduled and presented virtually to CO team and UNDP/Partner FPs in a weekly coordination meeting)

End-to-end demo: ***Registrar Admin*** *→* ***Wallet*** *→* ***Verifier*** *with Cardano testnet anchor.*

5.2 Testing & Feedback Documentation

Week 7–8 test plan, findings, and prioritized issues.

5.3 Feedback Integration Summary

List of changes shipped post-testing (UX copy, field mapping, claim guide, verifier display).

5.4 Pilot-Ready Solution

Stable issuer, wallet hand-off, and verifier, with revocation and status checks; docs + runbook.

5.5 Confirmed Pilot Partnership(s)

University of Sarajevo (UTIC + pilot faculties); UNDP Accelerator Lab Bosnia.

5.6 Stakeholder Alignment Matrix

Defined roles, responsibilities, and coordination plan with key partners.

Stakeholder / Partner	Role in the Pilot	Key Responsibilities	Level of Involvement (Lead / Support / Advisory)	Engagement Modality (e.g. MOU, Contract)	Focal Point / Contact Person
University of Sarajevo / UTIC	Issuer & Data Controller	Registrar ops, CSV schema, issuance/revocation	Lead	MoU/Letter	UTIC Lead
UNDP CO	Facilitator	Product mgmt, convening, user research	Lead/Support	Workplan	Kemal
Sala	Builder	Issuer service, verifier, wallet integration	Lead	Contract/MoU	Leap Sok

Emurgo Labs	Tech Partner	Testnet/endpoints, OID4VC guidance, reviews	Advisory	TA agreement	POC TBC

5.7 Budget Proposal (Pilot)

Provide a budget breakdown for pilot implementation. Include estimated cost, unit, quantity, and funding source.

Budget Line Item	Unit / Description	Quantity	Unit Cost (USD)	Total Cost (USD)	Funding Source (Confirmed / Pending)
TOTAL ESTIMATED COST				**[Insert Total]**	

5.8 Impact Measurement Framework

5.8.1 Contribution to relevant SDGs and indicators

(...)

5.8.2 Key Performance Indicators (KPIs)

- *Issuance ≥ **100** credentials (2 faculties).*
- *Verification ≥ **10** employers; ≥ **150** total verifications.*
- *p95 verify time ≤ **2s**; wallet claim success ≥ **90%**; median claim ≤ **2 min**.*
- *Verifier error rate < **1%**; revocation reflected ≤ **60s**.*
- *Satisfaction ≥ **80%** positive (students/employers).*
  (Aligned with earlier pilot goals and workplan cadence.)

5.8.3 Monitoring & Reporting Plan

a) ***Frequency:** weekly demo; bi-weekly KPI snapshot.*

b) ***Responsible:** Sala (tech), UNDP (user research), UTIC (registrar ops).*

c) ***Format:** short dashboard + notes; change log.*

d) ***Tools:** issue tracker, analytics, survey forms.*

5.8.4 Risks Assessment & Mitigation Plan

Risks	***Mitigation***
Scope creep	*lock MVP, weekly demos.*
Wallet friction	*1 wallet choice, 1-page claim guide, test low-end Android + iPhone.*
Testnet instability	*no "false valid", retry logic, cached reads.*
Data quality	*CSV validator/preview before issuing.*
Legal uncertainty	*run as complement; produce Canton memo.*
Adoption lag	*pre-confirm 8–10 employers; 30-sec verifier explainer. (Roadmap risks synthesized from workplan.)*

5.9 Post-Acceleration Roadmap & Sustainability Plan

5.9.1 Phased Implementation Timeline / Milestones

Short Term (0–6 months)	Medium Term (6–18 months)	Long Term (18+ months)
Expand to additional faculties; improve admin & verifier UX; finalize Canton memo.	Extend to other public universities; begin **IDDEEA wallet** integration track; consider Digital Public Good packaging.	National rollout with IDDEEA; multi-chain (Stellar status; FLock analytics) as needed.

5.9.2 Budget for Phased Implementation (post-acceleration)

5.9.3 Financial Sustainability Model

Institutional subscription/licensing to issuer/verifier modules; implementation/onboarding fees; optional support SLAs; potential DPG track for public components.

5.9.4 Partnership and Co-Financing Commitments

University (co-financing/hosting); UNDP (in-kind TA); Emurgo Labs (technical mentorship); employers' associations (testing).

5.9.5 Fundraising Plan

Grants (UNDP/UN, EU programmes), ecosystem funds (Cardano), institutional co-funding, and impact investors; timeline aligned with scale milestones.

5.11 Legal & Compliance Considerations

Data controller: University; Sala as processor. **No PII on-chain** (hash + status). Consent and retention per Canton guidance; accessibility (WCAG) and plain-language content; future alignment with **IDDEEA** digital wallet.

5.12 Visibility Assets / Communication Materials

1 slide "Why it matters: Weeks → Seconds; €0; Tamper-evident & revocable."
1 slide student journey (Issue → Wallet → Verify) + privacy note (no PII on-chain).
(Consistent with your kickoff deck.)

5.13 Pitch Deck

Concise and compelling presentation for stakeholders, funders, and partners. ((Please upload to your google drive folder)

5.14 Final Demo Presentation

An engaging walkthrough of the live solution and its implementation roadmap at the end of the program.

Deliverable Completion Table

Please indicate your intended completion week (or date) for each deliverable. You can revise this plan as the project evolves.

12-Week Delivery Plan (ordered)

Week	Focus & Decisions	Deliverables (what "done" means)	Owner(s)
W1	Kickoff & scope lock	1) **Challenge Definition** (1–2 pp), 2) **One-page architecture** (Issue → Wallet → Verify; **no PII on-chain**), 3) **Pilot partners confirmed** (UNSA/UTIC + 2 faculties).	Sala + UNDP + University
W2	Users, wallet, KPIs	1) **User Mapping & Testing Strategy** (personas, user stories, W7–8 test plan), 2) **Wallet decision** (OID4VC demo wallet; languages), 3) **Pilot KPIs** list.	Sala + UNDP + University
W3	First issuance path	1) **CSV/Excel schema agreed** (sample file with 10 fake rows), 2) **Issuer service stub** (accept CSV, validate), 3) **Cardano testnet** endpoints chosen (Preview/Pre-prod) + faucet.	Sala + EMURGO + University
W4	On-chain anchor (PoC)	**5.1 Solution Demo (PoC):** first **issuance** → **wallet receipt** → **on-chain hash anchor**; visible on explorer; short demo video.	Sala (+ EMURGO advisory)

W5	Verifier online	**Verifier portal (MVP):** QR input → shows **Valid/Revoked + issuer + credential + date**; no PII; uptime monitored.	Sala
W6	Admin online	**Registrar Admin (MVP):** CSV upload → preview → **Issue**/**Revoke**; audit log; basic RBAC.	Sala + University
W7	Pilot testing (round 1)	1) **Student wallet claim guide** (EN/BS, 1 page), 2) **Employer test pack** (30-sec explainer), 3) Run tests (students, 4–5 employers).	Sala + UNDP + University
W8	Feedback report	**5.2 Testing & Feedback Doc:** findings, metrics (claim success, verify p95), prioritized fixes; sign-off meeting notes.	Sala + UNDP
W9	Fixes & legal draft	1) **5.3 Feedback Integration Summary** (what shipped), 2) **Legal & Compliance memo (draft)** — controller/processor, consent text, retention, **no PII on-chain**.	Sala + University + UNDP
W10	Legal alignment final	**5.11 Legal & Compliance Mapping (final)**; update UI text/consents if needed; **Impact Measurement Framework** locked (reporting cadence).	Sala + University + UNDP
W11	E2E hardening	**Pilot-ready MVP:** issuer + wallet hand-off + verifier stitched; revocation reflected ≤60s; monitoring & runbook ready.	Sala
W12	Demo & comms	**Final Demo Presentation** (slides + live demo, 3 use cases), **Visibility assets** (student journey & “Why it matters”), **Pilot schedule** (faculties, volumes, dates).	Sala + UNDP + University

Appendix:

Section: Revoke

Building on the EBSI guidance and W3C specs, here's a "ready-to-adopt" revocation/status playbook the University can approve and we can implement.

1) Status model (what states exist)

- Use exactly three states: **Valid**, **Suspended**, **Revoked**. Suspended = temporary (e.g., missing coursework); Revoked = permanent (misconduct, withdrawal, duplicate, admin error). This mirrors EBSI's framing of status values. hub.ebsi.eu

2) Mechanism (how status is published)

- Adopt **W3C Status List 2021** (bitmap/bitstring) and the **EBSI profile** for it. It's privacy-preserving, space-efficient, and designed for high-volume issuers. W3C+1

- Host the university's Status List as a signed **Verifiable Credential** (VC) and update bits when a credential is suspended/revoked; verifiers read that list, not student data. hub.ebsi.eu

3) When to change status (policy triggers)

- **Suspend**: unpaid fees, incomplete thesis/exams, academic hold.

- **Revoke**: confirmed fraud, withdrawal, duplicate issuance, fatal errors that require re-issue.

- **Reinstate**: move from Suspended→Valid once academic/administrative conditions are met.
  (EBSI recommends choosing a status strategy that fits business/privacy requirements and short/long-lived VCs.) hub.ebsi.eu+1

4) Who can change status (governance)

- Define a **Registrar Status Authority** role (2-person approval or maker-checker) to change bits in the Status List and sign the updated list. Keep an **audit log** (who, what, when, why).

- Keys used to sign the list must be in **KMS/HSM** with rotation. (W3C VC security profiles + common KMS practice.) W3C

5) What goes on chain (privacy)

- Keep **all PII off-chain**. The blockchain (Cardano testnet) carries only a **hash/fingerprint** of the batch (or issuance event) and, optionally, a pointer to the status endpoint. Personal data stays in the wallet and university systems. W3C

6) How verifiers check status (runtime rules)

- Every verification checks two things:
    1. Signature of the diploma VC (issuer authenticity), and
    2. Current **status** via the Status List 2021 entry.
- DIF Presentation Exchange examples show including `credentialStatus` for revocation and `expirationDate` for expiry in verification requests. identity.foundation+1

7) Expiry vs. lifelong diplomas

- **Diplomas usually don't expire** (omit `validUntil`), but short-lived presentations **must** expire (minutes), and revocation is always checked at presentation time. W3C
- In the wallet share flow (OID4VP), bind each presentation to the **specific verifier** with a **nonce/audience** and short `exp` so screenshots/links can't be reused. openid.net

8) Performance & caching

- Status Lists are static files with signatures; serve them via CDN with **short TTL** (e.g., 60–120 s). If the list is unreachable, verifiers **fail closed** (show "Unable to check status")—never "Valid" by default. (Best practice derived from EBSI status guidance.) hub.ebsi.eu

9) Student UX & notices (plain language)

- Show **clear reasons** when status is not Valid:
    - Suspended: "Pending completion of [requirement]; contact registrar."
    - Revoked: "No longer valid; a replacement may be issued," with contact link.
- EBSI's policy docs also emphasize verifier authorization/purpose—restrict what employers can request by policy. hub.ebsi.eu

10) University SOP (1-page you can adopt)

1. Faculty/Registrar raises a status change with reason code.
2. Registrar Status Authority approves; system flips Status List bit(s) and republishes signed list.
3. Automated email to student (and, if needed, faculty admin) explaining status and remedy.
4. If reinstated, flip back to Valid and re-notify.
5. Quarterly key rotation; annual policy review.

11) Technical acceptance criteria (what we'll build)

- Status List 2021 endpoint exposed under university domain, signed VC per **EBSI profile**; list size and index management per defaults (e.g., 131,072 slots). hub.ebsi.eu
- Verifier portal **must** check: issuer signature, Status List bit, presentation expiry (OID4VP `exp`), and audience/nonce binding. openid.net
- Admin UI: **Suspend**, **Revoke**, **Reinstate** actions with reason codes and audit log.

12) FAQ you can share

- **Can the student still "see" a revoked diploma in their wallet?** Yes—wallets usually keep the record but it **verifies as Revoked**; Microsoft's Verified ID shows this

pattern. Microsoft Learn

- **Why Status List 2021?** It's the W3C mechanism designed to be **privacy-preserving and efficient**; EBSI profiles it for EU use. W3C+1

If you want, I'll turn this into a 2-page **Registrar Revocation Policy** (university letterhead) + a **tech checklist** for our devs so we can implement the Status List and admin actions next sprint.

What "revoking after issuance" actually means

- The copy in the student's wallet remains (it's their data).
- You flip its **status** to **Suspended** or **Revoked** in your **issuer-controlled status list**.
- From then on, any verifier that scans/checks it will see **Not valid**.

Best-practice pattern (W3C VC + Status List 2021)

Issue with a status pointer
Every diploma VC you issue should include:

```
"credentialStatus": {
  "id": "https://unisarajevo.ba/status/grad-2025#12345",
  "type": "StatusList2021Entry",
  "statusPurpose": "revocation",
  "statusListIndex": "12345",
  "statusListCredential": "https://unisarajevo.ba/status/grad-2025"
}
```

1.
    - This tells verifiers where to check current status.

2. **Maintain a signed Status List**
   Host `https://unisarajevo.ba/status/grad-2025` (a signed StatusList2021 VC).

    - Each credential maps to a **bit** (the index).

    - Bit = 0 → **Valid**; 1 → **Revoked** (you can also support **Suspended** via a second list or “suspension” purpose).

3. **Revoke or suspend** (after issuance)

    - In your **admin panel**, an authorized registrar flips the bit (maker-checker approval, audited).

    - Republish the signed list (same URL, new signature).

    - Result: all future verifications show **Revoked** (or **Suspended**).

4. **Verifier behavior (important)**

    - Verifier **must fetch** the list each time (short cache, e.g., 60–120s).

    - If status can’t be fetched, **fail closed**: show “Unable to check status” (never a false green).

5. **Presentations must expire quickly**

    - When a student shares, the wallet creates a **short-lived presentation** (e.g., 5–15 min, bound to the verifier with nonce/audience).
    - That way, a screenshot or old link can't bypass a later revocation.

"We anchored a hash on-chain—can we undo that?"

- You **don't remove chain data** (and you shouldn't need to).
- Because we put **no PII on-chain**, the old hash is harmless.
- Revocation lives **off-chain** in the status list; optionally, you can also anchor a **revocation evidence hash** on-chain for extra auditability, but it's not required.

Student experience (recommended UX)

- The wallet continues to show the diploma card, but with **Revoked** or **Suspended** status.
- If you **re-issue** a corrected credential (e.g., name fix), the student will see both cards; advise them to archive the old one.
- Send a polite notice when status changes, with next steps (e.g., "submit missing thesis chapter to reinstate").

University SOP (one page you can adopt)

1. Faculty requests status change with a **reason code** (e.g., "Missing exam", "Academic misconduct").
2. Registrar **Status Authority** reviews and approves (maker-checker).
3. System flips the bit in the Status List, republishes, logs **who/what/why/when**.
4. Notify student (and faculty admin).
5. If resolved, **reinstate** (flip bit back).
6. Quarterly **key rotation** for the status-signing key; annual policy review.

Edge cases & tips

- **Offline employers:** discourage purely offline checks; if offline, show “Unable to check status.”
- **Multiple statuses:** support both **suspension** (temporary) and **revocation** (permanent).
- **Data minimization:** revocation reason codes belong in your **internal log**, not in the credential or verifier screen.
- **Performance:** serve status lists via CDN with short TTL; lists scale to >100k entries per file.

What we’ll build for you (MVP)

- Status List 2021 endpoint under university domain (signed VC).
- Admin actions: **Suspend / Revoke / Reinstate** with reason codes + audit log.
- Verifier that checks: issuer signature → status bit → presentation expiry/nonce.
- Wallet flow that creates **short-lived, audience-bound** presentations.